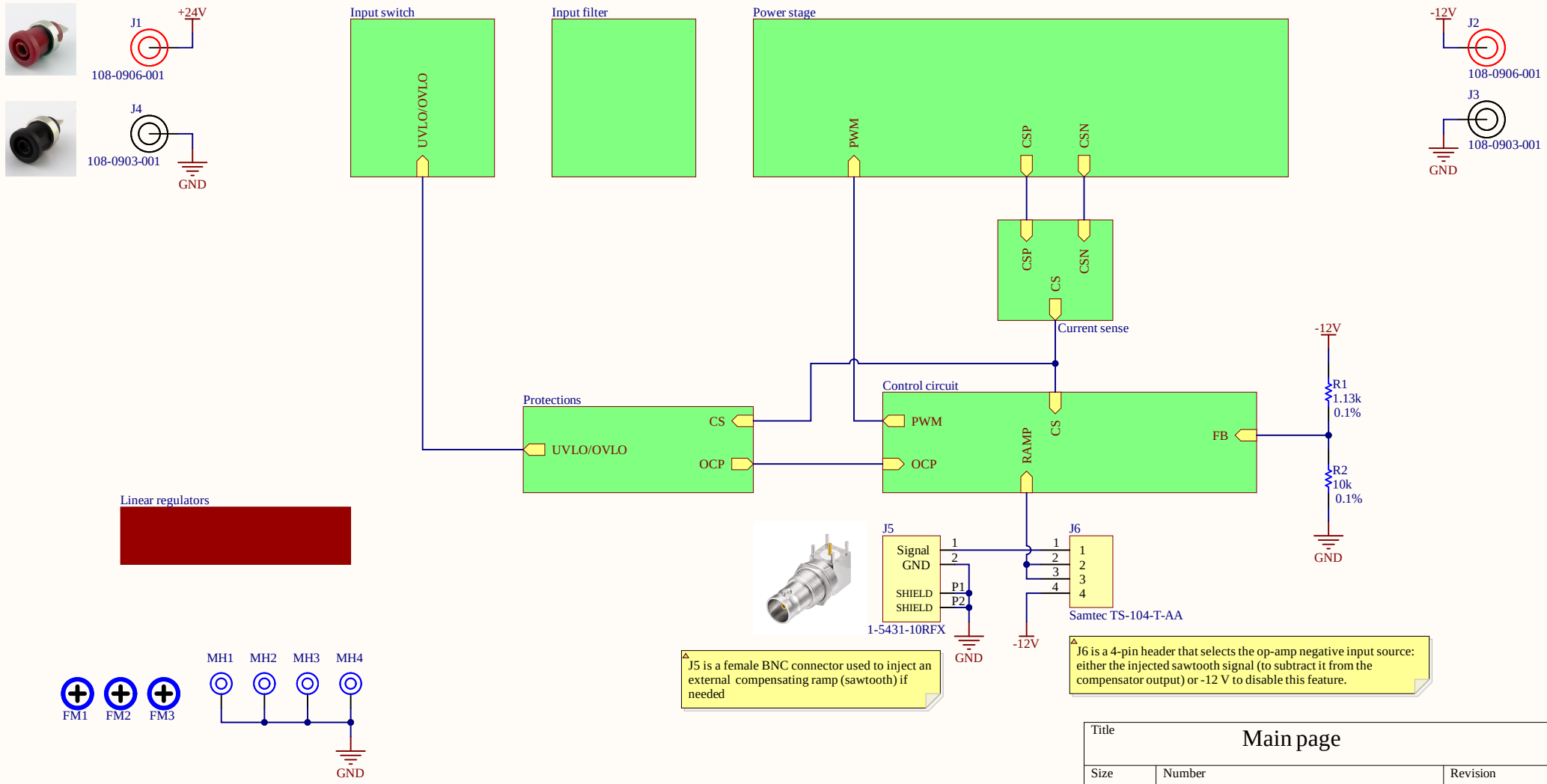
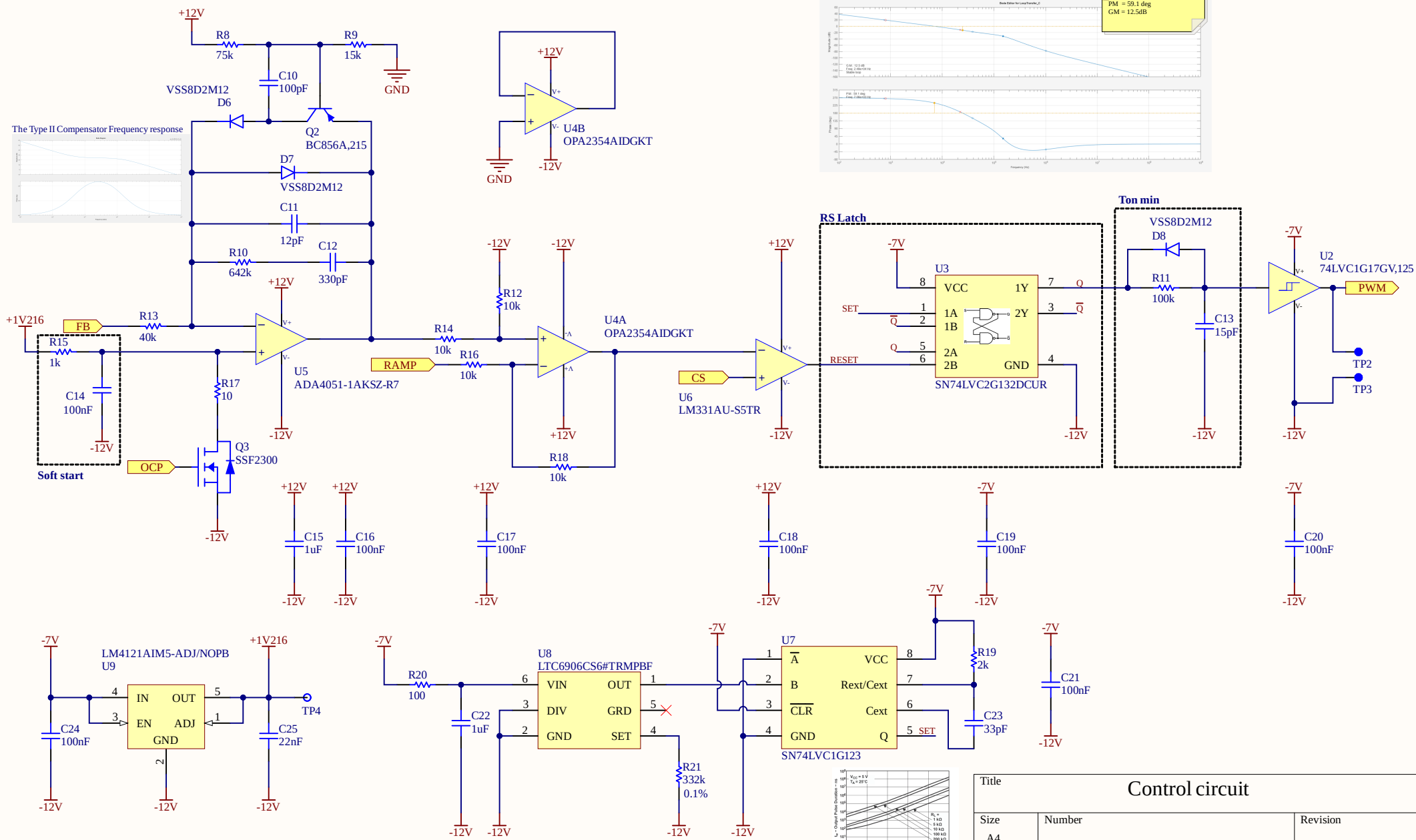


Inverting Buck-Boost



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A

A

B

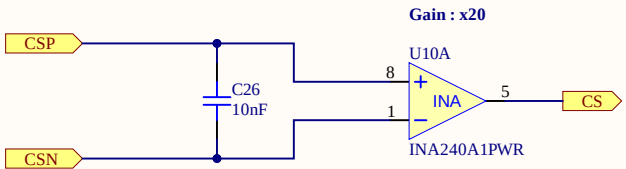
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C

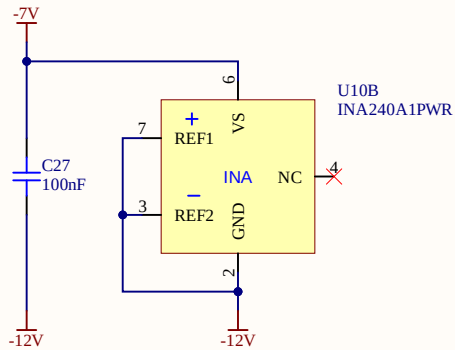
C

D

D



^A Total current gain = 0.015 x 20 = 0.3



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INPUT FILTER Design Notes

TOPOLOGY: Single-stage LC + parallel RC damping

SPECIFICATIONS:

$V_{in} = 24V$ | $f_s = 300kHz$ | $I_{out} = 3A$

FILTER CORNER FREQUENCY:

$f_c = f_s/20 = 300k/20 = 15kHz$
 $L_f = 47\mu H$ (selected) | $C_f = 2.2\mu F$
 $f_{c_actual} = 1/(2\pi\sqrt{L_f \times C_f}) = 15.65kHz$

ATTENUATION AT f_s :

$A = 40 \times \log_{10}(300k/15.65k) = 51.8dB$
Target = 17dB $\rightarrow$ Margin = 34.8dB ✓

CHARACTERISTIC IMPEDANCE:

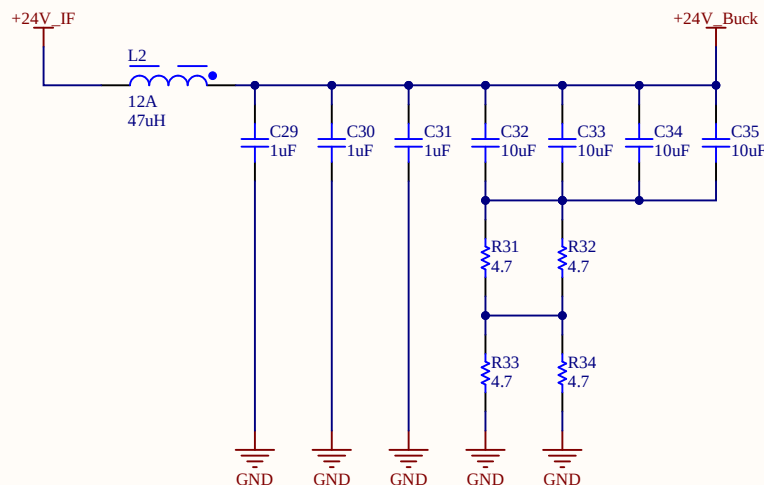
$Z_0 = \sqrt{L_f/C_f} = \sqrt{47\mu/2.2\mu} = 4.62\Omega$

DAMPING NETWORK:

$R_{damp} = Z_0 = 4.62\Omega \rightarrow 4.7\Omega$ selected
 $C_{damp} > 4 \times C_f = 4 \times 2.2\mu = 8.8\mu F$ selected
 $Q = R_{damp} \times \sqrt{C_f/L_f} = 4.7 \times 0.216 = 1.02$
NOTE: parallel damping topology

MIDDLEBROOK CRITERION:

$Z_{filter_out}(DC) \ll Z_{in_converter}$ ✓ (verified by MATLAB Graph below)

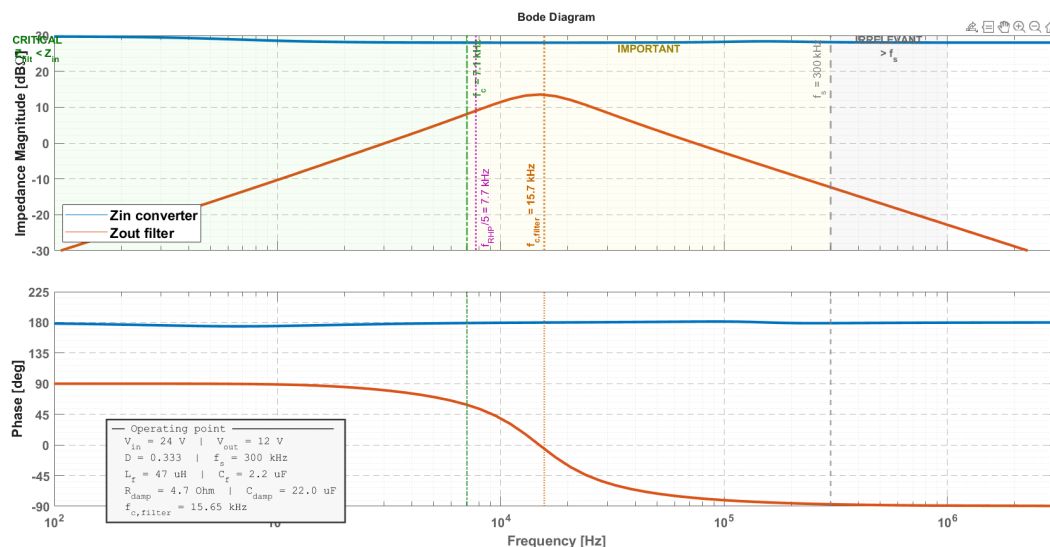
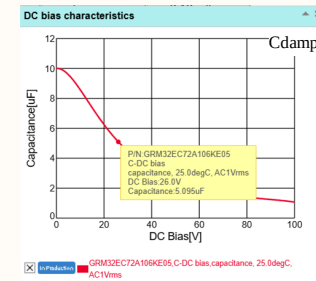
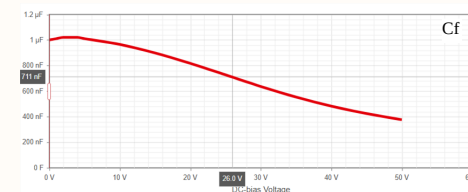


Cf selection:

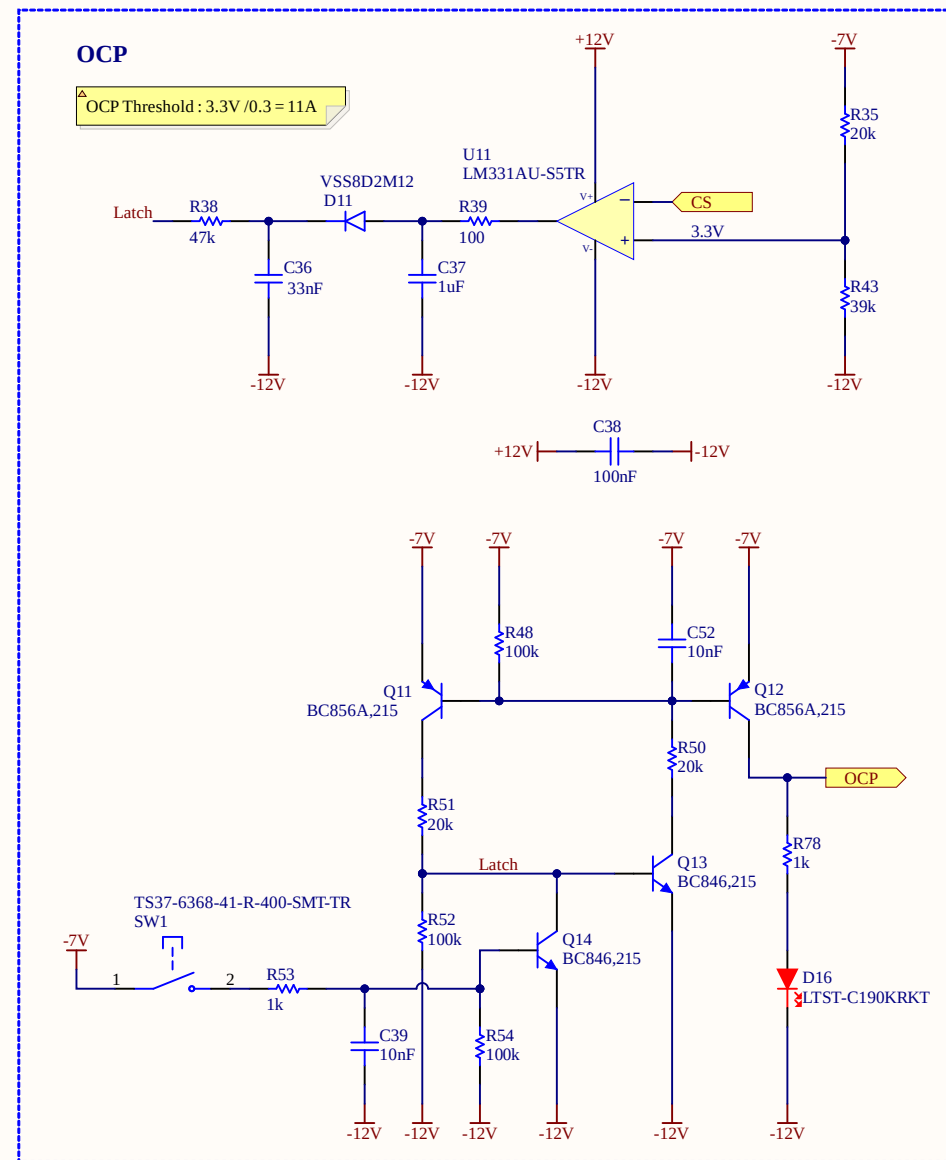
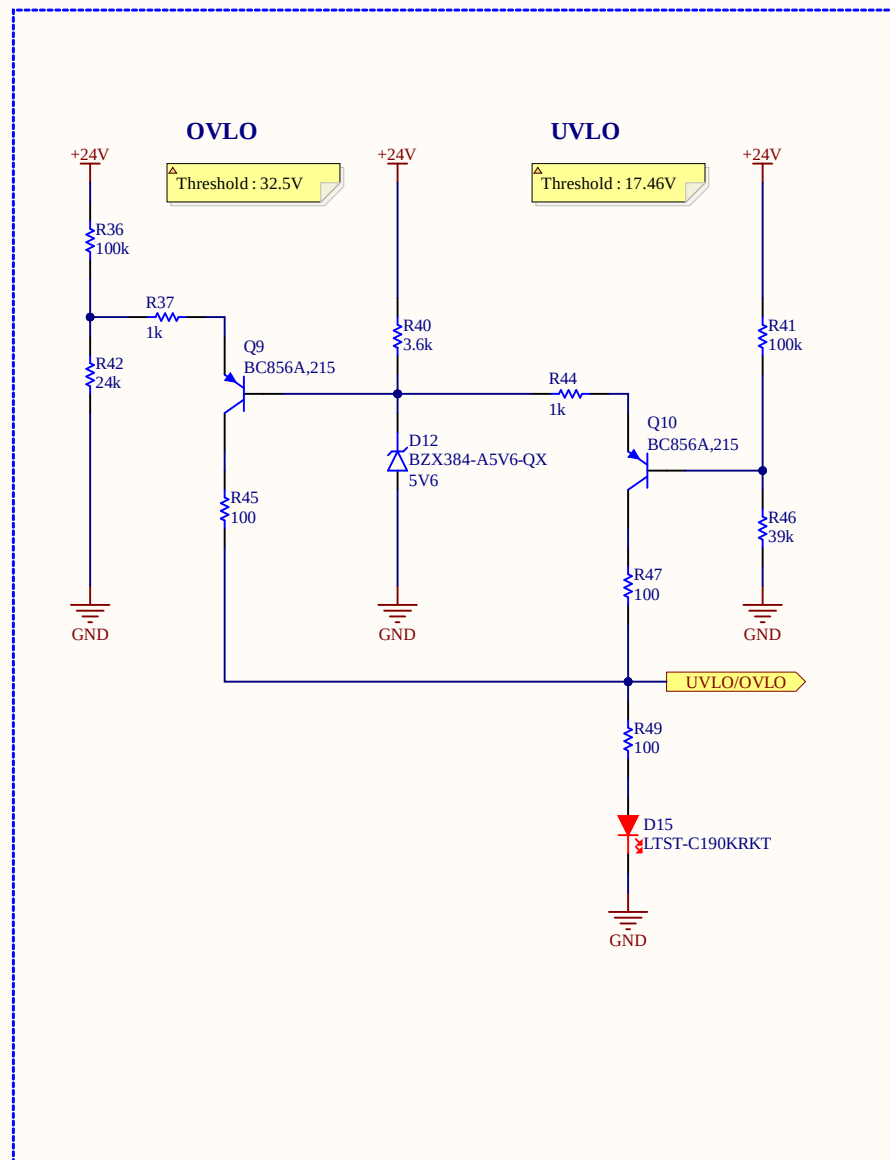
Nominal = 1uF per unit
Derated @ $V_{in} = 24V \rightarrow 0.73\mu F$ per unit
Units in parallel: 3
 $C_{f_effective} = 3 \times 0.73 = 2.19\mu F \approx 2.2\mu F$ ✓

Cdamp selection:

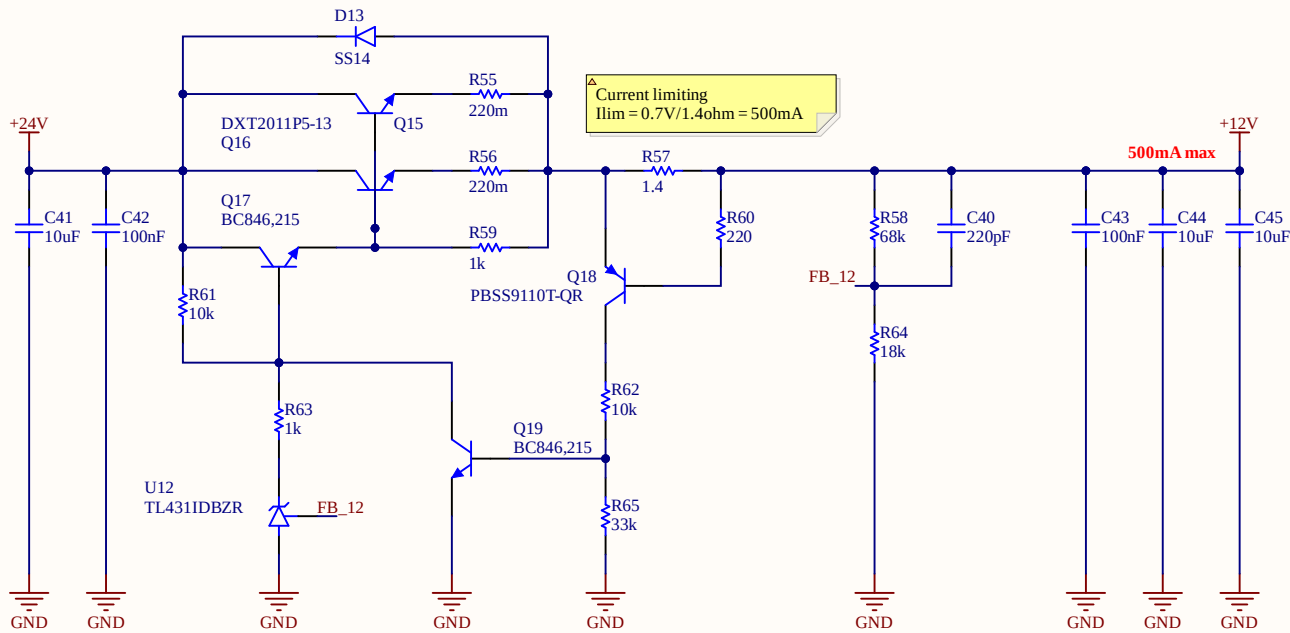
Nominal = 10uF per unit
Derated @ 24V $\rightarrow 5\mu F$ per unit
Required $C_{damp} = 8.8\mu F$
Units in parallel: 2
 $C_{damp_effective} = 4 \times 5 = 20\mu F > 8.8\mu F$ ✓



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△ To obtain the maximum possible current gain using Darlington pair it's important to ensure that the driver transistor has enough collector current to ensure acceptable h_{FE} . When run at very low collector current, most transistors will have a gain that's well below the quoted figure in the datasheet. The base-emitter resistor (R59 shown in the 12V regulator) is sized to ensure that the drive transistor has a collector current that's above the minimum. The resistor also helps to ensure a faster turn-off and minimises leakage current.

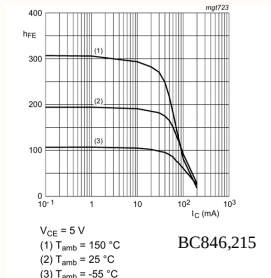
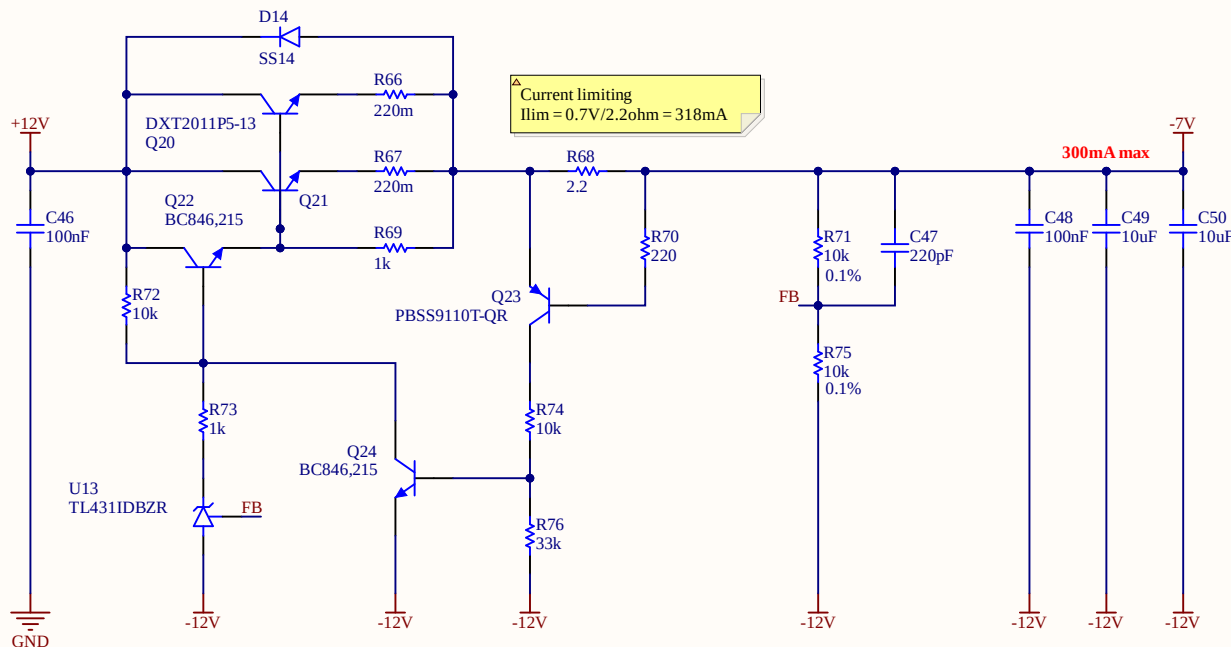
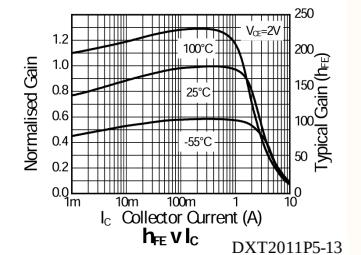


Fig. 2. Group A: DC current gain as a function of collector current; typical values

△ DXT2011PS can dissipate 3W if the device mounted with the exposed collector pad on 50mm x 50mm 2oz copper that is on a single-sided 1.6mm FR4 PCB

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